

Chapter 2. State of the Art

Over the past 10 years, the neural audio synthesis field was deeply changed with the main trend going from really slow autoregressive models towards streaming-causal architectures that can produce music in real time. This chapter is a guide to the main and extra works that shape the art of today, divided by the three main topics: (i) the architectural genealogy from WaveNet to BRAVE, (ii) methods of timbre transfer and voice conversion, (iii) datasets used to train and test these models.

The chapter finishes by clearly indicating the gaps in research that the present work is aiming to fill.

Literature Selection Methodology. The papers we have discussed have been chosen based on three factors: (a) direct architectural relevance to the streaming-causal Variational Autoencoders (VAEs) in this work, (b) methodological relevance to cross-modal timbre transfer, and (c) additional contributions to GAN stability and audio reconstruction quality passed on to the RAVE v2 architecture. The priority was given to highly ranked conference and journal venues (NeurIPS, ICML, ICLR, ISMIR, ICASSP) and arXiv preprints were used when they are the canonical reference for a widely used architecture. Literature searching was done from March to May, 2026. As a limitation of this survey, it should be mentioned that streaming neural audio synthesis being a very fast evolving field, some very recent architectures are mainly documented through preprints and open source implementations rather than peer-reviewed publications, and the canonical citation of several systems used in this project (most notably BRAVE) is an arXiv preprint.

2.1 Neural Audio Synthesis Architectures

Early Neural Synthesis Architectures

It was WaveNet (van den Oord et al., 2016), an autoregressive model factorizing the joint probability of the waveform into a product of conditional probabilities and generating audio sample by sample, that laid the groundwork for today's neural audio synthesis.

WaveNet that generated audio samples sequentially one at a time, achieved levels of naturalness in text-to-speech tasks that were never heard before subjectively, but live usage is unrealistic as performance is slow since every single audio sample needs the network to be run forward through it once.

NSynth (Engel et al., 2017) took up this model and applied it to musical notes by training a WaveNet autoencoder on over 300, 000 one-note sound samples. The authors demonstrated a 16-dimensional temporal embedding representation that can both capture the semantic timbral qualities of the sounds and allow morphing from one instrument identity to another.

NSynth still has the autoregressive production bottleneck of the original architecture (van den Oord et al., 2016) plus it only works on isolated notes as opposed to continuous performance.

Differentiable Signal Processing

Differentiable Digital Signal Processing (DDSP) (Engel et al., 2020) also gave us a fresh perspective. Instead of simply outputting raw audio samples, DDSP places conventional DSP modules, like harmonic additive synthesizers and filtered noise, directly into the computation graph and a neural network is trained to generate their control parameters. The authors suggest that the inductive bias of these synthesizers leads to significant reduction of data as well as model size requirements when compared to autoregressive models (Engel et al., 2020).

However, the price of this is that interpretability is exchanged for autonomy. DDSP is dependent on explicit fundamental-frequency (F0) and loudness signals in order to perform inference. In a live music performance, this entails an ongoing dependence on an external pitch-estimation module on the critical path.

Variational Autoencoders for Raw Audio: The RAVE Architecture

WaveNet and DDSP mainly focus on improving the reconstruction ability of the models by directly working on the raw waveform signals. Alternately, the Variational Autoencoder model (Kingma & Welling, 2013) taught us to learn structured latent variables that can be used both for accurate reconstruction and new data generation. RAVE (Caillon & Esling, 2021) changes this theory into practice for real-time audio.

Two-Stage Training Pipeline. RAVE first builds a VAE and then adversarially fine-tunes it. Phase 1 the multi-scale STFT loss is used for training the encoder-decoder pair that aims at minimizing the spectral reconstruction error (Engel et al., 2020; Caillon & Esling, 2021). Phase 2 the encoder is fixed and the decoder is trained to fool the multi-scale waveform discriminator (Goodfellow et al., 2014), thus enhancing the perceptual quality of the sound.

Multi-Band Decomposition for Real-Time Inference. To be able to run inference in mere real-time on typical CPU hardware, RAVE uses a 16-band frequency decomposition method based on a Pseudo-Quadrature Mirror Filter (PQMF) bank (Cruz-Roldán et al., 2002). This way the temporal dimension of the signal is approximately reduced by a factor of sixteen prior to the encoding step (Caillon & Esling, 2021).

Adversarial Stabilisation Techniques. Stability of the adversarial phase depends on the use of some extra techniques which have been adopted from the general generative-modelling literature: spectral normalisation of the discriminator (Miyato et al., 2018), multi-period discriminators taken from HiFi-GAN (Kong et al., 2020), and multi-scale spectral losses calculated over different STFT window sizes (Engel et al., 2020). These strategies together solidified RAVE's position as the neural timbre transfer architecture of choice for interactive applications.

Streaming-Causal Extension: BRAVE

Normal RAVE operates with non-causal convolutional layers and the end-to-end latency RAVE was measured at tens of millisecond range (Caillon & Esling, 2021), a latency that is very possible leadin to disruption in live musical performance.

BRAVE (Caspe et al., 2025) redesigns RAVE for streaming inference. The most important architectural modifications are: non-causal convolutions are swapped for causal ones, the compression factor of the encoder is minimized so as to limit the delay due to the accumulation of group delay, and the PQMF attenuation specification is relaxed to shorten the filter taps. Causal convolutions here are the main factor that allows this: by doing away the requirement of the future temporal context, they enable inference to run stepwise on streaming audio without buffering ahead of the current sample. Along with the kept convolutional state and a specialized C++ inference path, these modifications lower audio-to-audio latency to below 10 ms while maintaining the main qualitative characteristics of RAVE (Caspe et al., 2025).

SoundStream (Zeghidour et al., 2022), a neural codec made at the same time, independently showed that convolutional encoder-decoder topologies could be used for low-latency streaming when the padding schemes were designed with great care, thus providing architectural justification for the causal decisions of BRAVE. The downside, which BRAVE authors have explicitly acknowledged, is that the limited temporal context can negatively affect the pitch accuracy at extreme registers for melodic instruments (Caspe et al., 2025).

Synthesis: An Architectural Evolution Shaped by Latency

When looking at these four architectures consecutively, they form a distinct evolutionary path. WaveNet was the first to show that a neural network could generate waveforms directly and realistically, but it was only capable of working offline. DDSP made a breakthrough in allowing real-time inference by giving up the liberty of generating raw waveforms to the introduction of an inductive bias towards DSP-based synthesis, however, it needed an external pitch-estimation stage. RAVE eliminated the necessity of explicit pitch conditioning by discovering the latent space directly, but the use of non-causal context by it kept latency always higher than the level of perceptual immediacy for live usage. BRAVE manages to bring inference down to the sub-10 ms level by allowing an even smaller temporal context, thus it is the latest step in a decade-long journey away from offline waveform modelling and towards interactive musical instruments.

Project Architectural Choice. In fact, BRAVE is the only architecture among those reviewed that simultaneously meets the three major constraints of the project: implicit pitch modelling (eliminating the need for an external pitch tracker on the audio path), streaming inference and sub-10 ms end-to-end latency. Both WaveNet and NSynth are excluded by their autoregressive nature that is computationally expensive; DDSP is excluded by its reliance on external pitch tracking; standard RAVE is excluded by its non-causal context. Hence, the approach and assessment in Chapters 3 and 5 are geared towards BRAVE. Table 2.1 gives a brief overview of each architectural proposal from the perspective of live voice-controlled interaction.

Table 2.1. A concise comparison of the architectures analysed in Section 2.1. Latency is the end-to-end audio-to-audio delay as measured and reported by respective authors on their cited hardware configurations; entries without a directly measurable latency are qualitatively discussed.

Model	Real-time	Causal	Latency	Pitch conditioning	Latent representation	Main limitation for live voice-driven use
WaveNet (2016)	No	Yes (autoregressive)	Minutes per second of audio	Optional (linguistic features)	None (direct waveform)	Sample-by-sample generation prohibits live use
NSynth (2017)	No	Yes (autoregressive)	Minutes per second of audio	None	16-D temporal embedding	Restricted to isolated notes
DDSP (2020)	Yes	No	Real-time capable; constrained by external pitch tracker	Explicit F0 + loudness	Explicit DSP control parameters	External pitch tracker on critical path
TimbreTron (2018)	No	No	Offline (CQT + WaveNet vocoder)	Implicit via CQT	CQT spectrogram	Spectrogram-domain inference too slow
RAVE (2021)	Yes	No	~50 ms reported	Implicit	Learned VAE latent	Non-causal context limits perceptual immediacy

BRAVE (2025)	Yes	Yes	< 10 ms reported	Implicit	Learned VAE latent (causal)	Shorter temporal context can degrade pitch at extremes
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2.2 Timbre Transfer and Voice Conversion

The trajectory from Section 2.1 is about creating audio from a learned distribution. On the other hand, converting one sound to another in terms of timbre has a different literature and some of that is before RAVE. However, it still points to the cross-modal problems that this thesis is about.

Definition of Timbre

Timbre, the classical definition, is the perceptual characteristic that differentiates two sounds having the same pitch and loudness. Usually, it is talked about as a multidimensional term covering all the audible qualities that are not pitch or volume: spectral envelope, attack and decay characteristics, fine-grained temporal modulations, and the resonant properties of the source. When it comes to machine learning, timbre transfer is turning the sound features of a source signal into those of a target instrument while keeping the musical elements like pitch, rhythm, and dynamics.

Spectrogram-Domain Transfer

The first demonstrations of high-quality musical timbre transfer were given by TimbreTron (Huang et al., 2018), using CycleGAN, an image-to-image translation network, on constant-Q transform (CQT) spectrograms. Since CQT representations have the property of approximate pitch equivariance, they are particularly suitable for music. Phase-reconstruction, a problem which is always present in spectrogram-domain approaches, is solved by TimbreTron by attaching a conditional WaveNet vocoder to convert the transferred CQT to a waveform (Huang et al., 2018). The method produces very realistic piano-to-harpsichord and similar conversions but due to the WaveNet's offline inference time, it cannot be used live.

Voice Conversion

Voice conversion (VC) is about changing the speaker's identity while keeping the spoken content intact. AutoVC (Qian et al., 2019) proposed an autoencoder-only model where an information bottleneck separates speaker-dependent features from phonetic content, which makes zero-shot transfer possible even for target speakers not seen during training. Neural audio codecs like SoundStream (Zeghidour et al., 2022) take the encoder-decoder framework to the next level by performing general audio compression with residual vector quantisation, thereby allowing very high reconstruction quality at low bitrates.

The Voice-to-Instrument Cross-Modal Gap

Nevertheless, the difference between voice-to-voice and voice-to-instrument tasks is very significant. Voice production is characterised by the presence of a formant structure which is generated by the vocal tract, and the phonetic content which is communicated through rapid articulation. On the other hand, a plucked-string instrument like a guitar mainly derives its sound from mechanical excitation, resonance of the body, and an exponentially decaying sound envelope; such instruments don't produce formants and don't have phonetic content.

If a VAE is trained solely on guitars, then its latent space manifold is expected to be very poorly aligned with vocal features (Kingma & Welling, 2013; Caillon & Esling, 2021). Whether the encoder can position the voice within a meaningful region of that manifold and whether the decoder can generate a meaningful output based on that positioning is a question that this thesis aims to answer by means of the diagnostic methodology outlined in Chapter 3.

Synthesis: Timbre Transfer Trades Expressiveness against Real-Time Tractability

The above studies highlight the continuing dilemma in this area: on the one hand, you have expressiveness which is best served by unconstrained raw-waveform models such as TimbreTron; on the other hand, you have real-time tractability which is more in favour of encoder, decoder bottlenecks like AutoVC and RAVE. BRAVE, as a matter of fact, represents one extreme in this trade-off: it is willing to make use of strong inductive biases (causal convolutions, PQMF compression, fixed bottleneck) so that it can carry out inference in less than 10 ms.

What enables the model to be used in real-time also puts very severe limitations on its ability to intake and learn acoustical structural features of a source that is very different in nature, which is exactly the kind of scenario that the voice-to-guitar task in this thesis opens up.

2.3 Neural Audio Synthesis Datasets

The results of the RAVE-family models strongly correlate with the amount, variety, and training data uniformity. A duration from three to fifty hours of a single-source uninterrupted audio recording is recommended (Caillon & Esling, 2021).

GuitarSet

GuitarSet (Xi et al., 2018) contains acoustic guitar recordings of around three hours from six professional players in five musical styles. Besides the main microphone, each performance is recorded with a hexaphonic pickup that gives the isolation of each string. The annotations include pitch contours, beat positions, chord labels, and playing techniques. Although widely used as a music information retrieval benchmark, the generalization capacity of the models trained using this dataset alone, is limited due to its small duration and single-instrument nature.

Guitar-TECHS

Guitar-TECHS (Pedroza et al., 2024) is a multi-angled electric-guitar set of approximately 5.2 hours, covering six playing techniques: palm muting, harmonics, alternate picking, vibrato, string bends, and pinch harmonics. Each performance is recorded simultaneously through a direct-input (DI) channel, a close-miked amplifier, and stereo binaural microphones. The DI channel is preferred for RAVE training as it delivers a clean isolated signal without room acoustics.

IDMT-SMT-Guitar

IDMT-SMT-Guitar (Kehling et al., 2014) brings in the aspect of timbre diversity by featuring six guitars (five electric, one acoustic) that have been played by three musicians, and performances have been recorded using both microphone and DI capture paths. The dataset is divided into isolated notes (datasets 1, 2) and continuous performance excerpts (datasets 3, 4); only the latter is relevant for the current work.

Groove MIDI Dataset

Providing a percussive analog to the guitar collections is the Groove MIDI Dataset (Gillick et al., 2019): it contains approx. thirteen hours of acoustic drum-kit performances by professional drummers, with MIDI and

audio synchronisation across a wide range of styles and tempos. Besides comfortably exceeding the lower bound for RAVE training this also supports the secondary case study of this thesis.

Dataset Selection Rationale and Noise-Floor Heterogeneity

After selection for files that are compatible (mono, 16-bit, 44.1 kHz), the total time duration of the consolidated guitar corpus in this work (GuitarSet, the Guitar-TECHS DI channel, and the continuous-performance subsets of IDMT-SMT-Guitar) is approximately sixteen hours. This duration lies within the RAVE's recommended data window for training (Caillon & Esling, 2021) but also leads to a well-known risk: mixing recording paths leads to inconsistency in background noise floors. A quantitative analysis of the corpus in Chapter 5 reveals a 35.8 dB range in noise floor between the cleanest direct-input electric recordings and the noisiest microphone-captured acoustic recordings. This range is larger than the typical dynamics changes within a single performance and thus it might be encoded as a prominent latent feature by the encoder, biasing the learned manifold toward recording-method discrimination rather than solely timbral content.

2.4 Research Gap and Motivation

The literature review in this chapter presents a research landscape in which streaming neural audio synthesis has achieved the sub-10 ms latency needed for live musical interaction (Caspé et al., 2025); in which there are methods for timbre-transfer but they lose either real-time capability (TimbreTron) or controllability (DDSP); and in which the voice-to-instrument cross-modal task at this latency level remains nearly completely unexplored. BRAVE has shown voice-to-saxophone and beatbox-to-drums on smaller sets of data; however, it has not been tried for voice-to-guitar, and BRAVE's authors are quite clear about the fact that they recognize these questions as the ones this thesis tackles.

Most of the previous contributions that were devoted to the demonstration of success mostly only have a small part dedicated to failure and are not used as a continued focus for the work; failures in the models have been treated here as worthy objects of analysis. The three main issues addressed are as follows.

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- The pitch realization of plucked-string instruments For example, a sax usually can sustain a sound changing only the pressure of the breath whereas a guitar note is struck sharply and decays in an exponential manner. The interplay between such changes in the temporal dynamics and a streaming-causal encoder whose temporal context is restricted is one of the things not covered in the earlier BRAVE works. (Caspé et al., 2025)
- The sound difference between the voice and the guitar The voice's formants and the guitar's body resonance are found in completely different sections of the acoustic feature space. A question that is left open is whether a VAE manifold that is trained only on a guitar can be led to consider the voice as an input coming from the same distribution or if the voice is naturally an outlier of the trained manifold leading to the consequences mentioned in Chapter 5. This question has not been answered quantitatively so far.
- Training on multiple sources under very different recording conditions The question about what happens when one mixes the acoustic and electric guitar recordings, for which a noise floor spread of 35.8 dB was found in Section 2.3, with respect to BRAVE's latent space, which is the topic of this work, has not been answered by the previous works.

These studies serve as a foundation for the present research on several aspects of real-time voice instrument cross-modal conversion. They motivate the methodological decisions in the choice of BRAVE for deployment and the experimental design revealed in the next chapter.

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